Semantic-ID generative recommendation

Implemented M2 model · Amazon Reviews 2014 Beauty · simplified TIGER-style

A OFFLINE ITEM TOKENIZER

Product metadata
Title · brand
Categories · description

→

Frozen MiniLM
all-MiniLM-L6-v2
384-d normalized vectors

→

Hierarchical k-means
3 levels · branching ≤16
+ unique within-leaf suffix

→

Fixed item ↔ code table
12,068 items · [c1, c2, c3, u]
4 tokens per item · vocabulary 94

fixed item-to-code lookup

B TRAINABLE ENCODER–DECODER TRANSFORMER

History → code tokens
Last 20 items, in time order
BOS + codes + EOS · ≤82 tokens

→

Transformer encoder
Shared embedding + positions
2 layers · bidirectional attention

→

Transformer decoder
Shared embedding + positions
2 layers · causal + cross-attention

→

Next-token logits
Separate linear head · 128 → 94
Full-vocabulary scores

M2: d = 128 · 4 heads · FFN = 256 · dropout = 0.1

Input embeddings are shared; positions are sinusoidal.

BOS + previous code tokens
True prefix / generated prefix

C1 TRAINING

Next-item labels
[c1, c2, c3, u, EOS]
Using A's fixed code table

→

Token cross entropy
Full vocabulary · ignore PAD
Backpropagate to Transformer

Teacher forcing: decoder receives the true prefix.

Decoder input: [BOS, c1, c2, c3, u]

MiniLM and the item codes stay frozen.

C2 RECOMMENDATION

Catalog constraints
Valid-code prefix trie from A
Exclude ALL earlier seen items

→

Autoregressive beam search
Log-softmax → mask · width 200
Generate 4 code tokens + EOS

Rank by Σ log p, including EOS; no masked renormalization.

Code → catalog item lookup
Ranked unseen products · return top K

Solid arrows: model/data flow. Dashed arrow: fixed tokenizer lookup.

Warm-item retrieval over the training catalog